Lecture 2: Discrete Random Variables

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STAT-S520 Introduction to Statistics

Outline

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Basic Concepts

Discrete Random Variable

- Recall: random variable X is a function from sample space to value:

$$X : S \to \mathbb{R}$$

- A random variable X is discrete if its range $X(S)$ is countable. This is the possible values of X, namely $X(S)$, can be written as $\{x_1, \ldots, x_k\}$ for some finite k, or you can count all of its possible outcomes one-by-one (despite there might be infinite of them) as

$$\{x_1, x_2, x_3, \cdots\cdots\}$$

Basic Concepts

Example

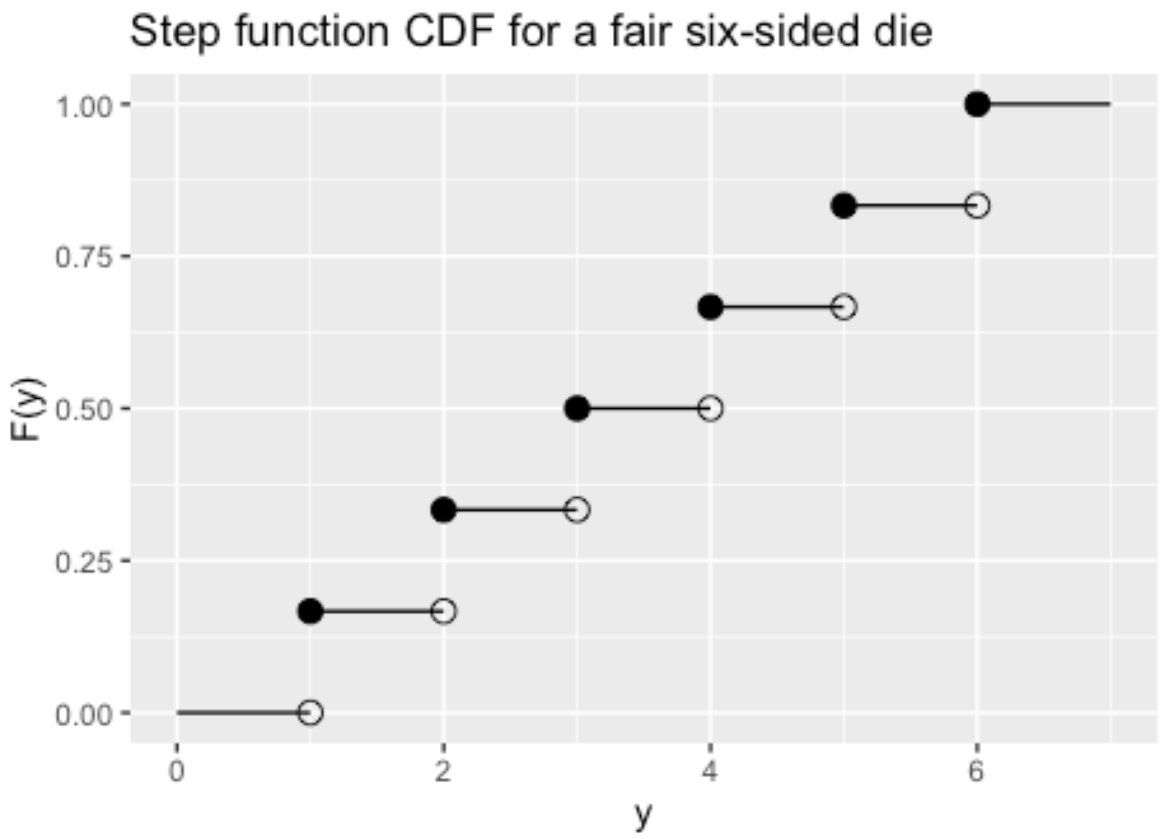
- Let X be the result of a fair die roll. Then the sample space $X(S) = \{1, 2, 3, 4, 5, 6\}$ is countable
- Flip n coins and let X be the number of heads. Then $X(S) = \{0, 1, 2, \ldots, n\}$ is countable
- Toss a coin and let X be 0.5 if it's heads and -0.5 if it's tails. Then $X(S) = \{-0.5, 0.5\}$ is countable
- Toss a coin repeatedly until you get a head, and let X be the total number of tosses. Then $X(S) = \{1, 2, 3, 4, \ldots\}$ is countable

Basic Concepts

Example

Let X be the result of a fair die roll. Then the outcomes in the sample space $S = \{1, 2, 3, 4, 5, 6\}$ are all equally likely. Then

$$P(X = x) = 1/6 \quad \text{for all } x \in X(S)$$



Basic Concepts

Example

- Let X be the result of a fair die roll
- Then the cdf of X is

$$F(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{6} & 1 \leq x < 2 \\ \frac{2}{6} & 2 \leq x < 3 \\ \frac{3}{6} & 3 \leq x < 4 \\ \frac{4}{6} & 4 \leq x < 5 \\ \frac{5}{6} & 5 \leq x < 6 \\ 1 & x \geq 6 \end{cases}.$$

- Using cdf to deal with the distribution of a discrete random variable is rather cumbersome

Basic Concepts

- The **probability mass function**, or the **pmf**, of a discrete function random variable is a function $f(x)$ such that

$$f(x) \equiv P(X = x)$$

  for all real numbers x.
- It gives the probability that a real number x is the outcome (even if that probability is zero)
- Since it's a probability, it has to be between 0 and 1
- Remember, for discrete random variables, F denotes the CDF, and f denotes the pmf.

Example

For a fair die roll, the pmf of the random variable involved is:

$$f(x) = \begin{cases} \frac{1}{6} & x \in \{1, 2, 3, 4, 5, 6\} \\ 0 & \text{otherwise.} \end{cases}$$

Basic Concepts

From pmf to cdf

- The CDF of a discrete random variable is a piece-wise constant function (step-function)
- To get the CDF at x from the pmf:

$$F(x) = \sum_{x_1 \leq x} f(x_1)$$

- Namely, add up all values of the pmf for x_1-values less than or equal to x
- Note that

$$F(\infty) = \sum_{x \in X(S)} f(x) = 1$$

- Namely, the pmf sums up to 1 (because probabilities have to add up to 1)

Basic Concepts

From cdf to pmf

- Let $F(x)$ be the cdf of a discrete random variable X
- Technically speaking, the pmf of X is given by

$$f(x) = P(X = x) = \underbrace{F(x)}_{P(X \leq x)} - \underbrace{\lim_{y \to x-} F(y)}_{P(X < x)}$$

- *If* there's a jump in the cdf at x, then $f(x)$, the value of the pmf at x, is the vertical size of the jump.
- If there's no jump in the CDF at x, then $f(x) = 0$

Basic Concepts

Example

A fair 4-sided die:

$$F(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{4} & 1 \leq x < 2 \\ \frac{2}{4} & 2 \leq x < 3 \\ \frac{3}{4} & 3 \leq x < 4 \\ 1 & x \geq 4 \end{cases}.$$

What's $f(3)$? What's $f(-7)$? What's $f(x)$ in general?

Basic Concepts

Example

A fair 4-sided die:

$$F(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{4} & 1 \leq x < 2 \\ \frac{2}{4} & 2 \leq x < 3 \\ \frac{3}{4} & 3 \leq x < 4 \\ 1 & x \geq 4 \end{cases}.$$

- At 3, the cdf jumps from $2/4$ to $3/4$. The size of the jump is $1/4$, so $f(3) = 0.25$.
- At -7, the cdf doesn't jump. So $f(-7) = 0$.
- The pmf for all x is

$$f(x) = \begin{cases} 1/4 & x \in \{1, 2, 3, 4\} \\ 0 & \text{otherwise.} \end{cases}$$

Basic Concepts

Bernoulli random variables: An essential example of discrete random random variables

- A random variable X follows a Bernoulli distribution if the range of X is $\{0, 1\}$.
- Bernoulli distribution is parameterized by the success probability $p = P(X = 1) \in [0, 1]$.
- $X \sim \mathrm{Bernoulli}(p)$.
- Bernoulli pmf is

$$P(X = 1) = p \quad \text{and} \quad P(X = 0) = 1 - p.$$

Outline

Basic Concepts

Expected Value

Binomial Distribution

Expected Value

- Let X be a discrete random variable with pmf f
- The **expected value** of a discrete random variable X, denoted by $E(X)$ or EX, is defined by

$$EX = \sum_{x \in X(S)} xP(X = x).$$

- This expected value is denoted $E(X)$ or EX or μ ("mu.")
- You can think of it as what you should "expect" the average value to be if the experiment is repeated a huge number of times.

Expected Value

- What is the average outcome you could expect for rolling a fair four-sided die?
- $S(X) = \{1, 2, 3, 4\}$. $P(X = 1) = 1/4$, $P(X = 2) = 1/4$, $P(X = 3) = 1/4$, $P(X = 4) = 1/4$.
- So we can expect the average of the outcome to be

$$\begin{aligned} EX &= 1 \cdot P(X = 1) + 2 \cdot P(X = 2) + 3 \cdot P(X = 3) + 4 \cdot P(X = 4) \\ &= 2.5 \end{aligned}$$

Expected Value

Example

Suppose an unfair four-sided die is weighted so that there's:

- a 20% chance of a one ($P(X = 1) = 0.2$);
- a 20% chance of a two ($P(X = 2) = 0.2$);
- a 30% chance of a three ($P(X = 3) = 0.3$);
- a 30% chance of a four ($P(X = 4) = 0.3$).

Then the expected value is

$$\begin{aligned} E(X) = \mu &= (1 \times 0.2) + (2 \times 0.2) + (3 \times 0.3) + (4 \times 0.3) \\ &= 0.2 + 0.4 + 0.9 + 1.2 \\ &= 2.7 \end{aligned}$$

This is a bit more than the 2.5 for the fair die. This makes sense: the big numbers are now a little more likely, and the small numbers a little less likely.

Expected Value

Important Properties of Expected Value

- If X is a random variable with $P(X = c) = 1$ for some constant c, then $E(X) = c \cdot P(X = c) = c$
- The expected value of $\phi(X)$

$$E[\phi(X)] = \sum_{x \in X(S)} \phi(x)\, f(x).$$

- A function ϕ of a discrete random variable X is also a discrete random variable.
- If a is a constant, then $E(aX) = a\, E(X)$:

$$E(aX) = \sum_{x \in X(S)} ax \cdot f(x) = a \underbrace{\sum_{x \in X(S)} x \cdot f(x)}_{E(X)}$$

- Let $Y = X^2$. Then Y is a random variable with

$$E(Y) = \sum_{x \in X(S)} x^2\, f(x).$$

Expected Value

Important Properties of Expected Value

- For any random variables X_1, X_2 defined on the same probability space (they are associated with the outcome of the same experiment),
$$E(X_1 + X_2) = EX_1 + EX_2.$$
- Tossing a fair coin once.
- $X_1 = 1$ if H and $X_1 = 0$ if T.
- $X_2 = 1$ if T and $X_2 = 0$ if H.
- $X_1 + X_2 = 1$ always, so $E(X_1 + X_2) = 1$.
- $EX_1 = EX_2 = 0.5$, so $E(X_1 + X_2) = EX_1 + EX_2$.
- Such property holds in general.

Expected Value

- Let X be a discrete random variable with expected value $\mu = EX$
- Consider the following function of X: $\phi(X) = (X - \mu)^2$
- Remember the key idea that **a function of a random variable is also a random variable**
- That means $\phi(X)$ too has an expected value:

$$E\phi(X) = E(X - \mu)^2 = \sum_{x \in X(S)} (x - \mu)^2 \cdot f(x)$$

- The quantity $E(X - \mu)^2$ is referred to as the **variance** of X and is denoted by

$$\mathrm{var}(X) = E(X - \mu)^2$$

- Instead of taking each outcome and multiplying by its probability ($E(X)$), here we take each outcome x, find $(x - \mu)^2$, and multiply by the outcome's probability $f(x)$. Then we add up the products ($\mathrm{var}(X)$).

Expected Value

- The variance is a measurement of the extent of deviation or fluctuation of a random variable:
- Tossing a fair coin once:

$$X_1 = \begin{cases} 1 & \mathrm{H} \\ 0 & \mathrm{T} \end{cases},$$

  and $X_2 = 0.5$ whatever the outcome is. Then X_1 and X_2 have the same expectation (the same expected average value). But X_1 has a larger deviation because of its randomness:

$$\mathrm{var}(X_1) = 0.5 > \mathrm{var}(X_2) = 0.$$

- Let X be a discrete random variable with variance $\mathrm{var}(X)$
- The **standard deviation** of X is the square root of the variance. It's denoted by sd or by the Greek letter σ:

$$\sigma \equiv \mathrm{sd}(X) = \sqrt{\mathrm{var}(X)}$$

- The sd has the interpretative advantage of being in the same units as the original random variable.

Expected Value

Important Properties of Variance

- Let X be a discrete random variable
- $\mathrm{var}(X) \geq 0$ for all discrete random variable X
- $\mathrm{var}(X) = E(X^2) - (EX)^2$ **(Most useful formula for variance!)**
- $E(X^2) \geq (EX)^2$.
- $\mathrm{var}(X + c) = \mathrm{var}(X)$ for any constant c
- $\mathrm{var}(cX) = c^2\mathrm{var}(X)$ for any constant c: the factor c augments (or shrinkages) the variance of the random variable by a factor of c^2.
- If X_1, X_2 are independent random variables defined on the same probability space, then $\mathrm{var}(X_1 + X_2) = \mathrm{var}(X_1) + \mathrm{var}(X_2)$

Expected Value

Example

- Let $X \sim \text{Bernoulli}(p)$ with $p \in [0, 1]$
- The expected value of X is

$$EX = \sum_{x \in \{0,1\}} xP(X = x) = 0 \times (1 - p) + 1 \times p = p$$

- Observe that $X \in \{0, 1\}$, then $X = X^2$, and hence

$$EX^2 = EX = p$$

- The variance of X is

$$\text{var}(X) = EX^2 - (EX)^2 = p - p^2 = p(1 - p)$$

Expected Value

Example

Consider the following two random variables:

- Game 1: Yichi bets on a fair coin toss. If it's heads, he wins \$1; if it's tails, he loses \$1. Then his profit X_1 has pmf

$$f_1(x) = 0.5, \quad x \in \{-1, 1\}$$

- Game 2: Yichi bets on a fair coin toss. If it's heads, he win \$100,000; if it's tails, he lose \$100,000. Then his profit X_2 has pmf

$$f_2(x) = 0.5, \quad x \in \{-100000, 100000\}$$

Both of these have expected value 0. Should Yichi be equally happy to play Game 1 and Game 2?

Expected Value

Example

For Game 1: The random variable X_1 has pmf

$$f_1(x) = 0.5, \quad x \in \{-1, 1\}$$

Then the variance is

$$\begin{aligned}
\mathrm{var}(X) &= E[(X_1 - \mu)^2] \\
&= \sum_{x \in \{-1,1\}} (x - \mu)^2 \cdot f_1(x) \\
&= [(1-0)^2 \cdot 0.5] + [(-1-0)^2 \cdot 0.5] \\
&= 0.5 + 0.5 \\
&= 1
\end{aligned}$$

Expected Value

Example

For Game 2: The random variable X_2 has pmf

$$f_2(x) = 0.5, \quad x \in \{-100000, 100000\}$$

Then the variance is

$$\begin{aligned}
\mathrm{var}(X_2) &= E(X_2 - \mu)^2 \\
&= \sum_{x \in \{-100000, 100000\}} (x - \mu)^2 \cdot f_2(x) \\
&= [(100000 - 0)^2 \cdot 0.5] + [(-100000 - 0)^2 \cdot 0.5] \\
&= (5 \times 10^9) + (5 \times 10^9) \\
&= 10^{10}
\end{aligned}$$

Game 2 has a much larger variance. It means the variation of the second Game 2 is much larger than Game 1 and Yichi should be happier to play Game 1 if he want to be safe and happier to play Game 2 if he would like to earn more at the cost of a higher risk.

Expected Value

Example

- Yichi has a four-sided die with probability 0.2 of a one, 0.2 of a two, 0.3 of a three, and 0.3 of a four.
- Previously we found the expected value to be 2.7.
- What's the variance?
- What's the standard deviation?

Expected Value

Example

- Suppose X is a discrete random variable with pmf

$$P(X=1)=P(X=2)=0.2, \quad P(X=3)=P(X=4)=0.3$$

- Then $EX = 2.7$
- We compute EX^2:

$$\begin{aligned} EX^2 &= \sum_{x\in\{1,2,3,4\}} x^2 P(X=x) \\ &= 1^2\times 0.2 + 2^2\times 0.2 + 3^2\times 0.3 + 4^2\times 0.3 \\ &= 8.5 \end{aligned}$$

- The variance is $\mathrm{var}(X) = EX^2 - (EX)^2 = 8.5 - 2.7^2 = 1.21$
- The standard deviation is $\mathrm{sd}(X) = \sqrt{\mathrm{var}(X)} = 1.1$

Outline

Basic Concepts

Expected Value

Binomial Distribution

Binomial Distribution

- Let $X_1, \ldots, X_n$ be independent $\mathrm{Bernoulli}(p)$ random variables
- Then the distribution of

$$X = X_1 + \ldots + X_n$$

  is defined to be the binomial distribution and is denoted by $X \sim \mathrm{Binomial}(n, p)$
- **pmf:** If $X \sim \mathrm{Binomial}(n, p)$ then

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x \in \{0, 1, \ldots, n\}$$

- **Expected value:**

$$EX = E(X_1 + \ldots + X_n) = EX_1 + \ldots + EX_n = np$$

- **Variance:** Since $X_1, \ldots, X_n$ are independent, then

$$\mathrm{var}(X) = \mathrm{var}(X_1 + \ldots + X_n) = \mathrm{var}(X_1) + \ldots + \mathrm{var}(X_n) = np(1-p)$$

Binomial Distribution

- The binomial is a good probability model to use for anything that's like "the number of heads in n independent spins of a coin, where each spin has probability p of coming up heads.",

Binomial Distribution

- Sampling with replacement and counting the number of draws that have some characteristic, where the probability of each draw can be regarded as a Bernoulli random variable and the multiple draws are treated independently
  - Put one black ball and one white ball in the urn and draw one ball each time with replacement. $X_i = 1$ if drawing a black ball at ith time, and $X = \sum_{i=1}^{n} X_i \sim \mathrm{Binomial}(n, p = 1/2)$ is the whole number of black balls.

Binomial Distribution

- Sampling without replacement *from an infinite population* and counting the number of draws that have some characteristic
  - A user watches n short videos recommended by a platform such as Instagram Reels or TikTok. Each video is randomly sampled from a huge population (henceforth, approximately infinite) is liked with probability p. Define $X_i = 1$ if the user likes the ith video. The total number of liked videos is $X = \sum_{i=1}^{n} X_i$. Since each X_i follows a Bernoulli distribution with parameter p and the trials are independent, we have $X \sim \mathrm{Binomial}(n, p)$.

Binomial Distribution

Take advantage of `R` when compute binomial probabilities

- It would be tedious to compute binomial probabilities by hand
- Instead use `R` commands reasonably:
- The R function `dbinom(x, n, p)` evaluates $f(x)$, the pmf of a Binomial(n, p) random variable at x
- The R function `pbinom(x, n, p)` evaluates $F(x)$, the cdf of a Binomial(n, p) random variable at x
- The order of the numbers in the parentheses is important!

Binomial Distribution

Example

Yichi gives a ten question true/false statistics test to a class of ten chimpanzees. The chimpanzees, who do not know any statistics, randomly guess true or false, independently for each question and independently of each other. A score of at least 8 out of 10 is required to pass the test.

1. One of the chimpanzees is named Lancelot. What is the probability that Lancelot passes the test?
2. What is the probability that at least one of the ten chimpanzees passes the test?

Binomial Distribution

Example

- *What is the probability that Lancelot passes the test?*
- Let X be a random variable representing the number of questions Lancelot gets right on a ten-question true/false test
- Lancelot gets the kth question right $X_k = 1$ and $X_k = 0$ otherwise.
- Then $X = X_1 + \cdots + X_{10} \sim \text{Binomial}(n = 10, p = 0.5)$.
- We need to find $P(X \geq 8) = 1 - P(X \leq 7) = 1 - F(7)$. We could do this by hand, but let's use R instead:

```
1 - pbinom(7, 10, 0.5)
[1] 0.0546875
```

Binomial Distribution

Example

- *What is the probability that at least one of the ten chimpanzees passes the test?*
- Let Y be a binomial random variable representing the number of chimpanzees who pass the test.
- There are ten chimps, so $n = 10$.
- The parameter p is the probability one single chimp would pass
- We found that on the previous slide: $p \approx 0.0547$.
- We need to find $P(Y \geq 1) = 1 - \underbrace{P(Y \leq 0)}_{F(0)} = 1 - \underbrace{P(Y = 0)}_{f(0)}$. In R:

```
p <- 1 - pbinom(7, 10, 0.5)
1 - dbinom(0, 10, p)
[1] 0.4301586
```

Binomial Distribution

Example

- Hotels that host conferences routinely overbook their rooms because some people who plan to attend conferences fail to show up.
- A common assumption is that 10 percent of the hotel rooms reserved by conference attendees will not be claimed.
- In contrast, only 4 percent of the persons who reserve hotel rooms for a science conference fail to claim them.
- Suppose that a certain hotel has 100 rooms. Believing that scientists behave like normal people, the hotel accepts 110 room reservations because $E(X_1 + \cdots + X_{110}) = 100$.
- What is the probability that the hotel will have to turn away statisticians who have reserved rooms?

Binomial Distribution

Example

- Let X_i be the random variable such that

$$X_i = \begin{cases} 1, & \text{if the } i\text{th scientist show up} \\ 0, & \text{if the } i\text{th scientist fail to show up} \end{cases}$$

  where $i = 1, 2, \ldots, 110$
- Then X_i's are random variables following $\mathrm{Bernoulli}(0.96)$
- We assume that $X_1, \ldots, X_{110}$ are independent
- Let $X = X_1 + \ldots + X_n$ where $n = 110$
- Then the event of interest has probability

$$\begin{aligned} P(X > 100) &= 1 - P(X \leq 100) \\ &= 1 \texttt{ - pbinom(100, 110, 0.96)} = 0.987 \end{aligned}$$